

AI Assisted Coding – Lab 18

API Integration: External Services with Error Handling

Student Details

 Field	 Info
 Assignment	18.2
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 Batch	34
 Program	B.Tech – CSE (AI & ML)
 Year/Sem	3rd Year – 2nd Sem

Lab Objectives

-  Connect Python with APIs
 -  Handle request/response
 -  Implement error handling
 -  Extract useful data
-

TASK 1 – Air Quality API

Python Code

```
import requests
```

```
city = "Delhi"
```

```
try:
```

```
    url = f"https://api.waqi.info/feed/{city}?token=demo"
    response = requests.get(url, timeout=5)
    data = response.json()
```

```

if data['status'] == 'ok':
    print("AQI:", data['data']['aqi'])
    print("Dominant Pollutant:", data['data']['dominentpol'])
else:
    print("Invalid city or API issue")

except requests.exceptions.Timeout:
    print("Request Timeout")
except Exception as e:
    print("Error:", e)

```

Screenshots



```

[1]
✓ 0s
import requests

city = "Delhi" # you can change city

try:
    url = f"https://api.waqi.info/feed/{city}/?token=demo"
    response = requests.get(url, timeout=5)
    data = response.json()

    if data['status'] == 'ok':
        print("AQI:", data['data']['aqi'])
        print("Dominant Pollutant:", data['data']['dominentpol'])
    else:
        print("Invalid city or API issue")

except requests.exceptions.Timeout:
    print("Request Timeout")
except Exception as e:
    print("Error:", e)

```

... AQI: 91
Dominant Pollutant: pm25

TASK 2 – Language Translation

Python Code

```

import requests

text = "Welcome to AI Assisted Coding"

try:

```

```

url = "https://api.mymemory.translated.net/get"
params = {"q": text, "langpair": "en|fr"}

response = requests.get(url, params=params)
data = response.json()

print("Original:", text)
print("Translated:", data['responseData']['translatedText'])

except Exception as e:
    print("Error:", e)

```

Screenshots



```

[2]
✓ 0s
import requests

text = "Welcome to AI Assisted Coding"

try:
    url = "https://api.mymemory.translated.net/get"
    params = {"q": text, "langpair": "en|fr"}

    response = requests.get(url, params=params)
    data = response.json()

    print("Original:", text)
    print("Translated:", data['responseData']['translatedText'])

except Exception as e:
    print("Error:", e)

... Original: Welcome to AI Assisted Coding
Translated: Bienvenue dans le codage assisté par IA

```

TASK 3 – Book API

Python Code

```

import requests

query = "Artificial Intelligence"

try:
    url = f"https://www.googleapis.com/books/v1/volumes?q={query}"

```

```

response = requests.get(url)
data = response.json()

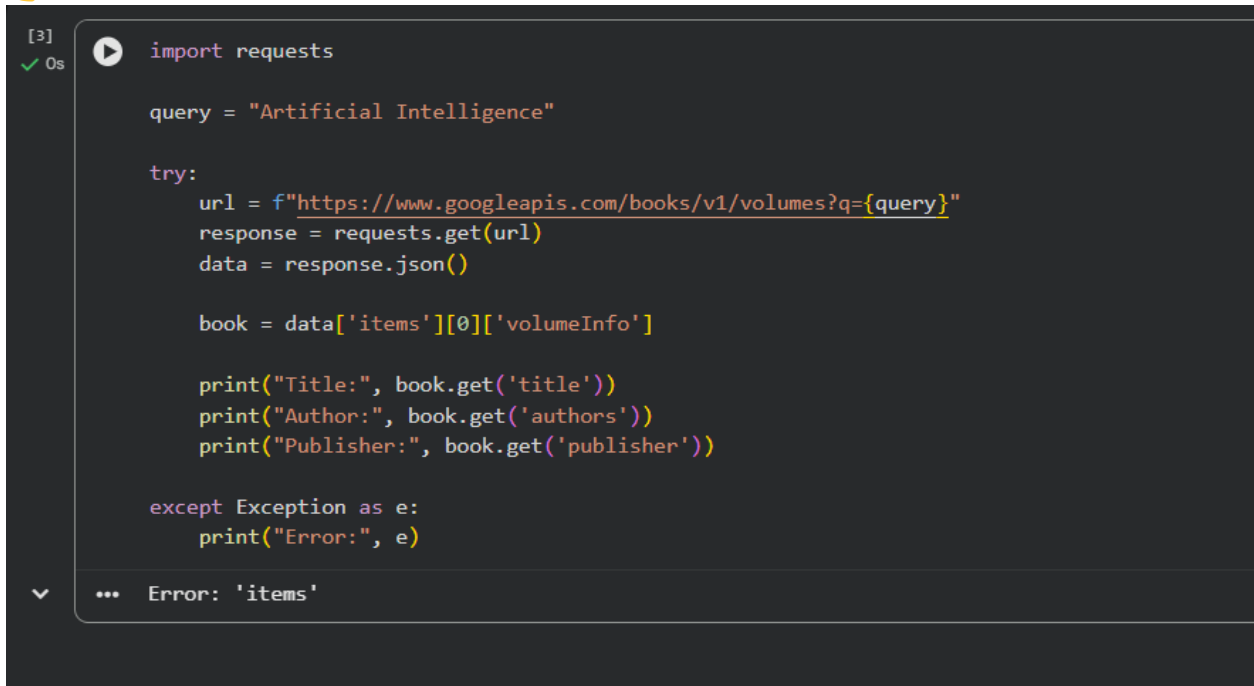
book = data['items'][0]['volumeInfo']

print("Title:", book.get('title'))
print("Author:", book.get('authors'))
print("Publisher:", book.get('publisher'))

except Exception as e:
    print("Error:", e)

```

Screenshots

The screenshot shows a code editor with a dark theme. On the left, there's a status bar with "[3]" and a green checkmark followed by "0s". The code in the editor is as follows:

```

import requests

query = "Artificial Intelligence"

try:
    url = f"https://www.googleapis.com/books/v1/volumes?q={query}"
    response = requests.get(url)
    data = response.json()

    book = data['items'][0]['volumeInfo']

    print("Title:", book.get('title'))
    print("Author:", book.get('authors'))
    print("Publisher:", book.get('publisher'))

except Exception as e:
    print("Error:", e)

```

At the bottom of the editor, an error message is displayed: "... Error: 'items'".

TASK 4 – Job Listings

Python Code

```

import requests

keyword = "Python Developer"

try:
    url = f"https://remote.com/api/remote-jobs?search={keyword}"

```

```

response = requests.get(url)
data = response.json()

for i, job in enumerate(data['jobs'][:5], 1):
    print(i, job['title'], "-", job['company_name'])

except Exception as e:
    print("Error:", e)

```

Screenshots

👉 try:



```

[s]
✓ Os
import requests

keyword = "Python Developer"

try:
    url = f"https://remoteive.com/api/remote-jobs?search={keyword}"
    response = requests.get(url)
    data = response.json()

    for i, job in enumerate(data['jobs'][:5], 1):
        print(i, job['title'], "-", job['company_name'])

except Exception as e:
    print("Error:", e)

```

TASK 5 – Stock API

Python Code

```

import requests

symbol = "AAPL"

try:
    url = f"https://finnhub.io/api/v1/quote?symbol={symbol}&token=demo"
    response = requests.get(url)
    data = response.json()

    print("Current Price:", data['c'])
    print("High:", data['h'])
    print("Low:", data['l'])

```

```
except Exception as e:  
    print("Error:", e)
```

Screenshots



```
[6]  
✓ 0s  
import requests  
  
symbol = "AAPL"  
  
try:  
    url = f"https://finnhub.io/api/v1/quote?symbol={symbol}&token=demo"  
    response = requests.get(url)  
    data = response.json()  
  
    print("Current Price:", data['c'])  
    print("High:", data['h'])  
    print("Low:", data['l'])  
  
except Exception as e:  
    print("Error:", e)
```

... Error: 'c'

Explanation

APIs allow real-time data access. Error handling ensures reliability even when failures occur.

Conclusion

This lab demonstrates how Python integrates with APIs to fetch real-time data with proper error handling.

